INSTITUTE OF INFORMATION TECHNOLOGY · UNIVERSITY OF DHAKA

SE801 · SOFTWARE PROJECT LAB 3 · FINAL DEFENSE · OCTOBER 2026

Terra-Audit

An AI-Assisted Digital MRV Platform for AWD Rice Irrigation and Agricultural Land Management Carbon Credits

Turns free satellite evidence into a reviewed, verifier-ready carbon report, so every tonne of CO_2e can be traced to its source.

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METHODOLOGIES
Verra VM0051 · VM0042
+ VMD0054 v1.1 leakage adapter

STACK
Next.js · FastAPI · PostgreSQL
Python worker · Google Earth Engine

LIVE APP
terra-audit.vercel.app

SOURCE CODE
github.com/knah1d/terra-audit

A · THE CHALLENGE

1 Introduction & Background

Flooded rice fields release **methane**, a strong greenhouse gas. **Alternate Wetting and Drying (AWD)** drains the field at least twice per season and cuts that methane while keeping yields. To earn carbon credits, a project must **prove** the reduction to an independent verifier.

1.00 → 0.55 Daily methane factor, flooded → AWD: **45% lower**
IPCC 2019 Table 5.12, used by Verra VM0051

- **MRV** Measurement, Reporting & Verification of a carbon claim
- **SAR** Satellite radar that sees through cloud (Sentinel-1)
- **VVB** Accredited third party that verifies carbon claims

2 Problem Statement

Practice is hard to prove
Irrigation logs are incomplete; sensors on every smallholder plot are too costly.

Numbers lose their evidence
Spreadsheets drop the factors, methodology version and history behind a result.

Verifiers need a trail
Every tonne of CO_2e must trace back to its evidence and its rule.

Gap: satellite tools and carbon spreadsheets live apart, so reviewers rebuild the link by hand.

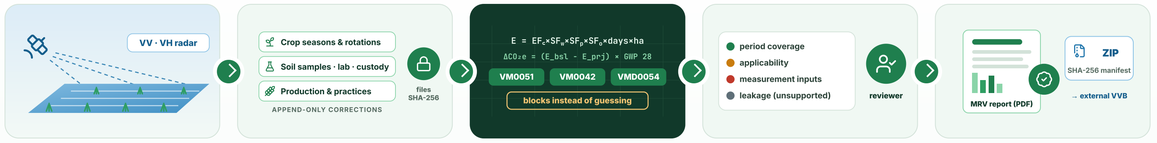
3 Research Objectives

- **Detect AWD from space** — Free Sentinel-1 radar, labelled as estimates
- **Rule-based engines** — Scoped VM0051 & VM0042 with visible steps
- **Cropland evidence** — Practices, production, soil, lab & custody
- **Readiness & leakage** — Gaps shown, never counted as met or zero
- **Frozen versions** — Each result keeps its exact evidence
- **Internal review** — A second person approves every result
- **AI as an aid only** — ML & LLM never change a value
- **Multi-user platform** — Roles, background jobs, history & export

B · OUR APPROACH

4 Graphical Abstract

FROM SATELLITE EVIDENCE TO A VERIFIER-READY REPORT



01 · MEASURE
Sentinel-1 radar
Earth Engine time series show when each field was flooded and drained, without field sensors.

02 · COLLECT
One evidence record
Seasons, soil, production and files stay linked to the field; corrections are append-only.

03 · CALCULATE
Deterministic engines
Verra equations with cited IPCC factors; every intermediate value is kept.

04 · CHECK & REVIEW
Readiness & review
Checks block incomplete work; a second person approves or requests changes.

05 · REPORT
Verifier-ready package
MRV report (PDF) plus an evidence ZIP with a SHA-256 manifest.

2 drydowns → 2.12 tCO_2e / ha
Live run: Sentinel-1 found 2 AWD drydowns in an 84-day season; VM0051 gives the **net** reduction after uncertainty and N_2O deductions.

Advisory AI: explains, never decides
An ML cross-check and cited LLM explanations go to a human reviewer. No AI can change a value or create credits.

5 Materials & Methodology: End-to-End Workflow

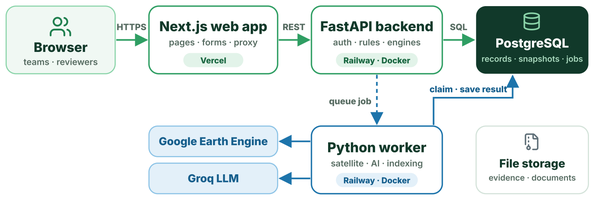
EVERY STEP IS SAVED WITH ITS EVIDENCE

01 · Register field — Draw or upload the boundary; area, district and type are auto-detected. — FIELD TEAM
02 · Record season — Crop, dates, practices and files; corrections are appended. — FIELD TEAM
03 · Satellite analysis — A background job pulls Sentinel-1, smooths it and counts drydowns. — AUTOMATIC
04 · Calculate — The engine runs on a frozen snapshot; gates and readiness are checked. — ONE CLICK
05 · Review — A different person records findings and approves or asks for changes. — REVIEWER
06 · MRV report — PDF report plus a SHA-256-hashed evidence package for the VVB. — PROJECT LEAD

FROZEN AT COMMIT (IMMUTABLE SNAPSHOT)
Field boundary & area · Seasons & engine inputs · Methodology bundle & factors · Readiness result

VERSION & REVIEW STATES
`draft` → `ready_for_review` → `submitted` → `in_review` → `internally_approved`

6 System Architecture



Durable job queue: the API only queues slow satellite and AI work, so pages never hang. Each job goes to exactly one worker and is recovered if that worker dies.

7 AI Authority Boundary

CODE DECIDES · DETERMINISTIC
Evidence → Validation → VM0051 / VM0042 engine → Result & readiness

AI EXPLAINS · ADVISORY
Scoped context → LLM → Validator → Draft → Human reviewer

Bounded — Sees only the active methodology, evidence and calculation.
Validated — 14 checks: citations exist, numbers match, no "certified".
Fail closed — One repair try, then fail; nothing partial is saved.

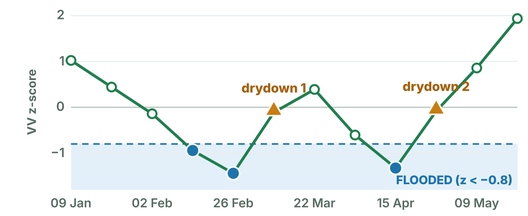
No AI component can change a value, mark a rule compliant, or create credits.

C · EVIDENCE IT WORKS

8 Results & Analysis

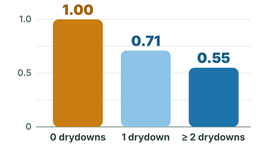
FROM THE LIVE APP & TEST SUITE

A · Sentinel-1 detects the drydowns (live run)



● flooded ▲ drydown (radar rises > 1.2σ right after a flooded date). 12 scenes, Jan–May 2026 → **2 AWD events, 84-day season**.

B · Why drydowns matter



Methane scaling factor by water regime (IPCC Table 5.12).

C · Worked VM0051 result



tCO_2e · 1 ha · 84 days · 2 drydowns · 100 kg N/ha.

D · ML cross-check

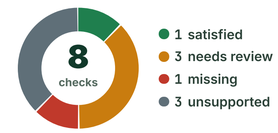
72.7%	71.1%	0.719	0.754
PRECISION	RECALL	F1	AUC

AWD plots found on held-out data: 64 / 90

863 labelled plots · 688 train · 172 test · 5×5 CV 66.3% ± 4.2%

Random forest on Sentinel-1 + Satellite Embedding. Punjab labels, so **advisory only**.

E · Readiness gate



VM0051 field example. Missing: AWD events & season length. Unsupported rules (e.g. leakage) stay visible, so the version stays `draft`.

F · Software verification

184 backend tests · 0 failed

- 4 / 4 frontend tests
- Production build
- TypeScript · 0 lint errors

TC-01	500,000 ha → QA3 size gate blocks it	PASS
TC-03	Other organization → 404	PASS
TC-04	Edited evidence → PDF from snapshot	PASS
TC-06	AI says "certified" → rejected	PASS

Not claimed: AWD field ground truth · Browser E2E tests · 20-case AI evaluation

Software verification ≠ VVB verification ≠ Verra issuance

D · WHAT IT MEANS

9 Discussion

- **Free radar can surface AWD** without field sensors, but results stay estimates: a field can drain and re-flood between passes.
- **Blocking beats guessing:** gates stop unsupported claims; tests caught a wrong emission factor (1.4 → 0.85).
- **Freezing gives reproducibility:** a report rebuilt later shows the same numbers and evidence.
- **ML is a second opinion:** about 71% of unseen plots are right with out-of-region labels, so it stays advisory.

10 Conclusion

Terra-Audit shows that the evidence behind an agricultural carbon claim can be **collected, calculated under VM0051 and VM0042, and reviewed in one system**, traceable from field to final number. Anything the methodology cannot support is blocked, not reported.

Satellite evidence · 2 Verra engines · Frozen versions

11 Future Scope

- **P0** Fix CI ordering, close endpoint authorization gaps, resolve the additionality/readiness conflict.
- **P1** **Field ground-truth campaign** for AWD in Bangladesh; tamper-evident evidence manifest; finish the AI evaluation.
- **P2** Versioned migrations; Eq. 38 uncertainty for large projects; remaining VM0051 terms with VVB review.

12 · KEY REFERENCES

[1] Verra, *VM0051 v1.1: Improved Management in Rice Production Systems.*
[2] Verra, *VM0042 v2.2: Improved Agricultural Land Management* (11 Jun 2026 corrections).
[3] Verra, *VMD0054 v1.1: Leakage from the Displacement of Agricultural Activities.*
[4] IPCC, *2019 Refinement to the 2006 Guidelines*, Vol. 4, Ch. 5, 10, 11.
[5] IPCC, *AR5: The Physical Science Basis* (2013), GWP100 values.
[6] ESA/Copernicus, *Sentinel-1 SAR User Guide*; Google Earth Engine (S1_GRD).
[7] Lampayan et al., "Adoption and economics of AWD," *Field Crops Res.* 170, 2015.
[8] Savitzky & Golay, *Anal. Chem.* 36(8), 1964; Breiman, "Random Forests," 2001.
[9] Microsoft, *rice-irrigation-mapping-s1s2* dataset (Punjab AWD labels).

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